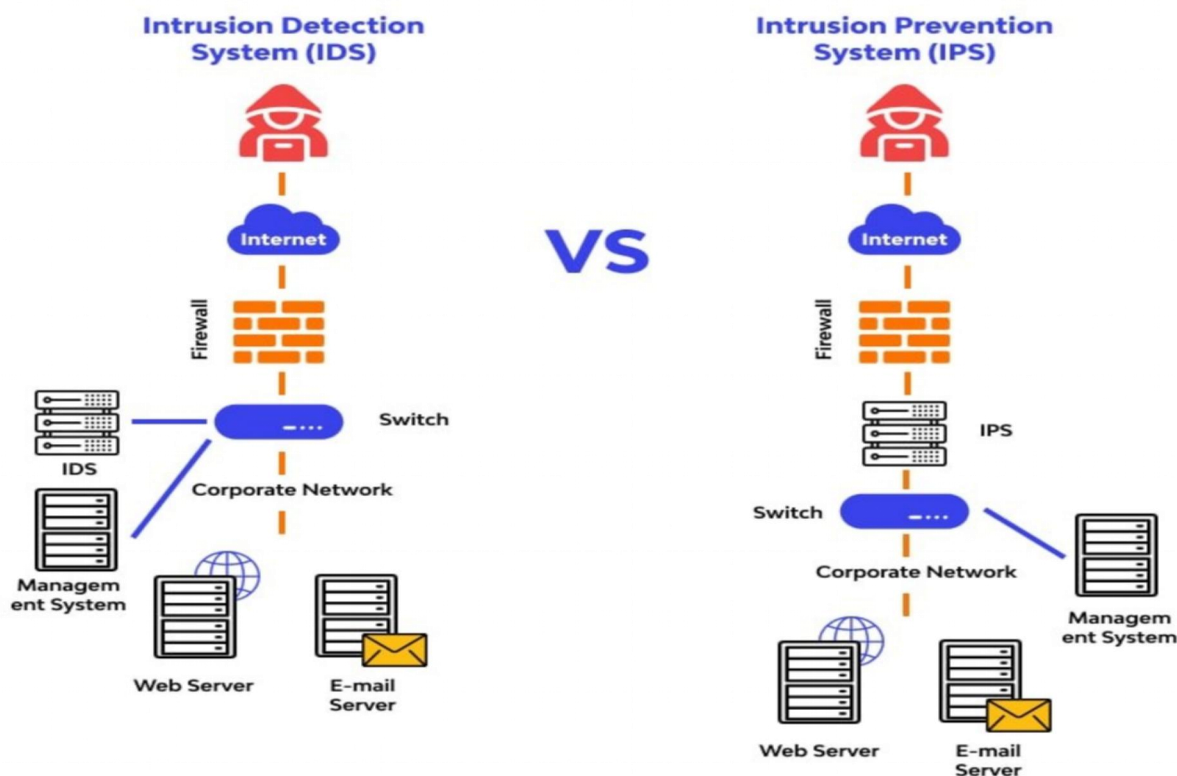




Activity 8

OPNSense Intrusion Prevention System

Be able to fully configure an “OPNSense Intrusion Prevention System” in a virtual environment using Oracle VirtualBox.



Activity Procedure(s)

Configure the network adapter settings of all virtual machines as listed below. Once completed, run/start virtual machines in the following order (OPNSense, Kali Linux, Metasploitable, and Ubuntu).

Virtual Machine Name	Network Adapter Settings
OPNSense	Adapter 1 (NAT)
	Adapter 2 (Host-only Ethernet Adapter #2)
	Adapter 3 (Host-only Ethernet Adapter #3)
	Adapter 4 (Host-only EthernetAdapter #4)
Kali Linux	Adapter 1 (Host-only Ethernet Adapter #2)
Metasploitable	Adapter 1 (Host-only Ethernet Adapter #3)
Ubuntu	Adapter 1 (Host-only EthernetAdapter #4)



Run/Start virtual machine(s) simultaneously (OPNSense, Kali Linux, and Ubuntu) from Oracle VirtualBox manager. In your Kali Linux virtual machine, open a web browser and access the OPNSense Web Management portal to configure OPNSense Intrusion Prevention services (Services > Intrusion Detection > Administration).

Next, create a Suricata Rule(s) that is intended to be implemented in your OPNSense Intrusion Prevention services (refer to your course manual for the “Suricata Rules Creation and Implementation Guide”) with the following IDPS Rules requirements listed below:

1. Denial-of-Service Attack Detection and Restriction Rules (SYN Flood, FIN Flood, RST Flood)
2. Web Attack Detection and Restriction Rules (SQL Injection)
3. Adult Site Detection and Restriction Rules (minimum of 5 sites)

After successful creation and implementation of Suricata rules in OPNSense, perform the necessary attack simulation to verify the functionality of the IDPS rules. Provide the necessary proof as needed.

Submission Note (Individual Activity)

Use file name convention (LASTNAME_CTAINASL_SECTION_TERM_AY_Activity8.pdf).

Submit/upload Softcopy (PDF file) in MS Teams

Submit a PRINTED activity rubric.



ACTIVITY DOCUMENTATION

Group Name

The Gantt gang

Monday, May 18, 2026

Members Surname, First Name MI. (Alphabetical)

Lalu, Mark Julius E..

Click or tap here to enter text.

Click or tap here to enter text.

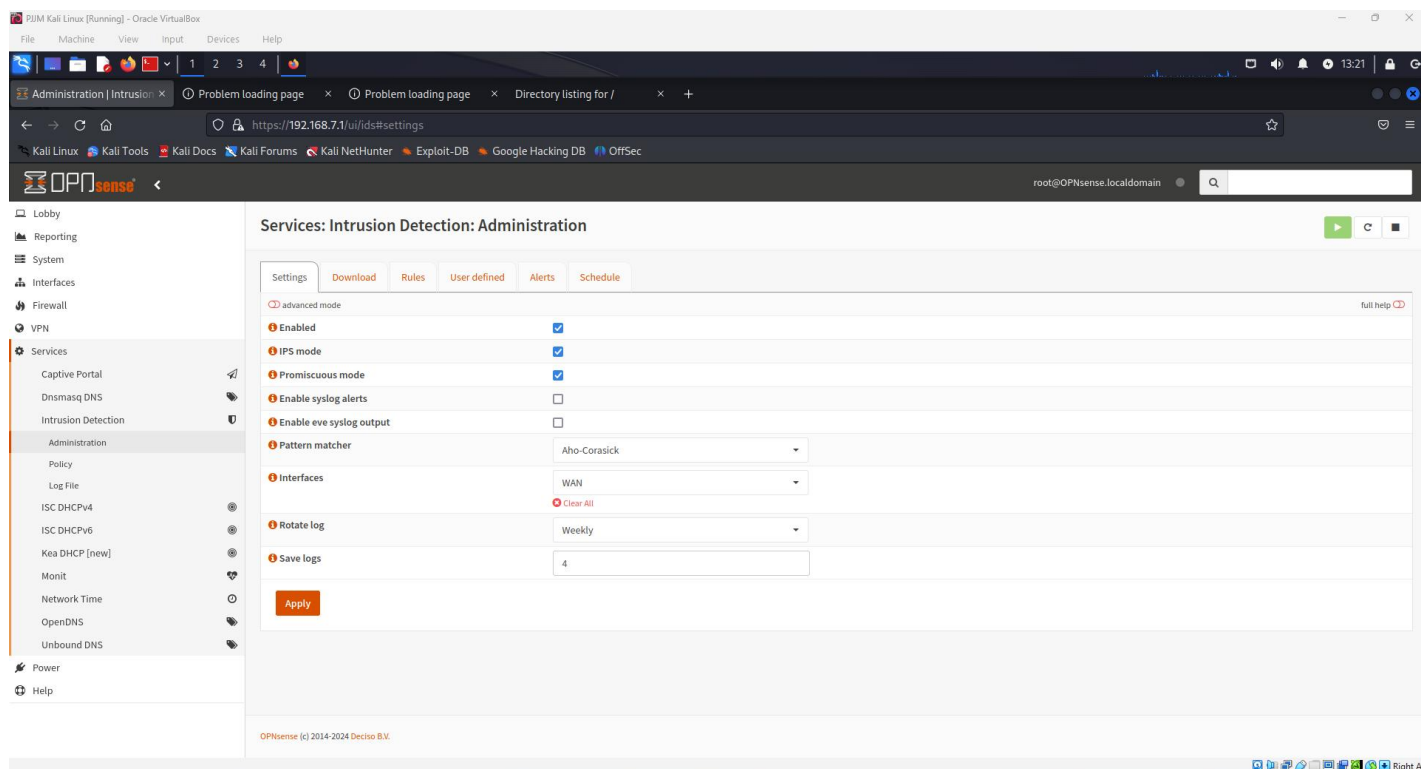
Click or tap here to enter text.

Click or tap here to enter text.

Instruction(s): Provide the appropriate screenshot/screen capture of your workstation.

OPNSense Intrusion Prevention Settings

Display here a screenshot of your OPNSense Intrusion Prevention service settings configuration.





OPNSense Intrusion Prevention Ruleset

Display here a screenshot of your custom ruleset implemented in OPNSense Intrusion Prevention service (Download Tab).

The screenshot shows the OPNSense Administration interface. The left sidebar contains a menu with options: Lobby, Reporting, System, Interfaces, Firewall, VPN, Services, Captive Portal, Dnsmasq DNS, Intrusion Detection, Administration, Policy, Log File, ISC DHCPv4, ISC DHCPv6, Kea DHCP [new], Monit, Network Time, OpenDNS, Unbound DNS, Power, and Help. The main content area is titled "Services: Intrusion Detection: Administration" and has tabs for Settings, Download, Rules, User defined, Alerts, and Schedule. The "Download" tab is active, showing a list of rulesets. The "Rulesets" section has buttons for "Enable selected" and "Disable selected". A search bar is present. The table lists various rulesets with columns for checkboxes, names, status, and actions. The "Download & Update Rules" button is at the bottom.

Rulesets	Enable selected	Disable selected
☐ ET open/emerging-web_specific_apps	not installed	x
☐ ET open/emerging-worm	not installed	x
☐ ET open/threatview_CS_c2	not installed	x
☐ ET open/tor	not installed	x
☐ INFOSEC Rules/INFOSEC Rules	2026/05/04 1:01	✓
☐ OPNSense-App-detect/file-transfer	not installed	x
☐ OPNSense-App-detect/mail	not installed	x
☐ OPNSense-App-detect/media-streaming	not installed	x
☐ OPNSense-App-detect/messaging	not installed	x
☐ OPNSense-App-detect/social-networking	2026/05/03 23:27	✓
☐ OPNSense-App-detect/test	2026/05/03 23:27	✓
☐ OPNSense-App-detect/uncategorized	not installed	x

Display here a screenshot of your enabled ruleset implementation in OPNSense Intrusion Prevention service (Rules Tab).

The screenshot shows the OPNSense Administration interface. The left sidebar is the same as the previous screenshot. The main content area is titled "Services: Intrusion Detection: Administration" and has tabs for Settings, Download, Rules, User defined, Alerts, and Schedule. The "Rules" tab is active, showing a list of rules. The "Filters" section has a search bar and a dropdown menu. The table lists rules with columns for checkboxes, sid, Action, Source, ClassType, Message, and Info / Enabled. The "Alert" checkbox is checked. The "Apply" button is at the bottom.

Filters	sid	Action	Source	ClassType	Message	Info / Enabled
☐	51000059	alert	opnsense.social_media.rules	social-media	OPN_Social_Media - Xing - Related TLS SNI (xing.com)	✓
☐	56000003	alert	INFOSEC.rules	uncategorized	OPN_Uncategorized - PornHub - DNS request for po...	✓
☐	56000004	alert	INFOSEC.rules	uncategorized	OPN_Uncategorized - PornHub - Related URL (porn...	✓
☐	56000005	alert	INFOSEC.rules	uncategorized	OPN_Uncategorized - PornHub - Related TLS SNI (p...	✓





OPNSense Intrusion Prevention Alerts

Display here a screenshot of your OPNSense Intrusion Prevention Alerts service based on the result of performing various attack simulation.

The screenshot shows the OPNSense Administration web interface. The left sidebar contains a navigation menu with options like Lobby, Reporting, System, Interfaces, Firewall, VPN, Services, and Power. The 'Services' section is expanded, showing 'Intrusion Detection' and 'Administration'. The main content area is titled 'Services: Intrusion Detection: Administration' and has tabs for Settings, Download, Rules, User defined, Alerts, and Schedule. The 'Alerts' tab is active, displaying a table of intrusion alerts. The table has columns for Timestamp, SID, Action, Interface, Source, Port, Destination, Port, Alert, and Info. The alerts listed are all 'allowed' actions on the 'wan' interface, originating from 10.0.2.15 and destined for 8.8.8.8 on port 53. The alert messages are 'OPN_Uncategorized - PornHub - DNS request for pornhub.com'. The footer of the interface shows 'OPNsense (c) 2014-2024 Deciso B.V.' and system status icons.

Timestamp	SID	Action	Interface	Source	Port	Destination	Port	Alert	Info
2026-05-04T01:14:11.519521+0800	56000003	allowed	wan	10.0.2.15	8510	8.8.8.8	53	OPN_Uncategorized - PornHub - DNS request for pornhub.com	
2026-05-04T01:14:11.269309+0800	56000003	allowed	wan	10.0.2.15	32408	8.8.8.8	53	OPN_Uncategorized - PornHub - DNS request for pornhub.com	
2026-05-04T01:14:11.269309+0800	56000003	allowed	wan	10.0.2.15	32408	8.8.8.8	53	OPN_Uncategorized - PornHub - DNS request for pornhub.com	
2026-05-04T01:12:40.982526+0800	56000003	allowed	wan	10.0.2.15	35476	8.8.8.8	53	OPN_Uncategorized - PornHub - DNS request for pornhub.com	
2026-05-04T01:12:40.982526+0800	56000003	allowed	wan	10.0.2.15	35476	8.8.8.8	53	OPN_Uncategorized - PornHub - DNS request for pornhub.com	
2026-05-04T01:11:13.926813+0800	56000003	allowed	wan	10.0.2.15	26169	8.8.8.8	53	OPN_Uncategorized - PornHub - DNS request for pornhub.com	
2026-05-04T01:11:13.650777+0800	56000003	allowed	wan	10.0.2.15	2288	8.8.8.8	53	OPN_Uncategorized - PornHub - DNS request for pornhub.com	

Activity Reflection

What is your personal realization based on the performed activity?

I have realized that this is somewhat similar to the previous activity but for this one you don't block the connection to the internet itself but rather, to specific websites that are inappropriate for certain environments like adult websites and other malicious attacks. You could also utilize the suricata rules to avoid certain cyber attacks online like DDoS and SQL Injection if used correctly.





ACTIVITY RUBRICS

Group Name The Gantt gang

Monday, May 18, 2026

Members Surname, First Name MI. (Alphabetical)

Lalu, Mark Julius E.

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Click or tap here to enter text.

Criteria	Activity Rubrics Self-Paced Activity 8 OPNSense Intrusion Prevention					Points
	Not Attempted (0 points)	Beginning (1 point)	Developing (2 points)	Proficient (3 points)	Exemplary (4 points)	
Use of Tools & Techniques	No attempt to use relevant tool(s).	Incorrect or unsuitable tool(s) selected.	Tool(s) used is/are somewhat suitable but not optimal.	Selected appropriate tool(s) with minor mismatches to the scenario.	Selected the most appropriate tool(s) for the task based on scenario.	
IPS/IDS Configuration	No attempt to configure IPS/IDS.	IPS not enabled or misconfigured; minimal use of Suricata rulesets.	Basic IPS setup; defaults mostly unchanged; limited logging.	IPS enabled with proper rulesets; logging and alerts partially configured.	Suricata IDS/IPS fully configured; interfaces, rulesets, and logging set appropriately.	
Custom Suricata Rule Creation	No attempt to create custom rules.	Rules missing, invalid, or copied without understanding.	Rules created with basic structure but lack specificity or show errors	Rules are mostly accurate and functional; minor logic or syntax errors.	Custom rules are syntactically correct, effective, and demonstrate understanding.	
Testing & Validation	No attempt to test and validate created rules.	Test or results invalid/inconclusive.	Minimal testing conducted; unsure if rules are triggering correctly.	Test scenarios run with partial success; some alerts observed.	Test traffic properly generated and detected by custom rules; alerts verified.	
Documentation	No attempt to provide documentation of findings.	Poor documentation of findings lacks structure or critical details.	Basic report provided with significant omissions or unclear explanations.	Detailed report provided; minor gaps in methods or findings.	Comprehensive report including methods, and findings.	
Total Score and Fe-edback					TOTAL POINTS EARNED (20 max points)	
☐ Exemplary	18-20	Mastery of IPS setup and rule creation with strong testing/reporting.				
☐ Proficient	15-17	Good technical execution; minor issues in configuration or documentation				
☐ Developing	12-14	Basic completion; lacks depth or clarity in parts of the task.				
☐ Beginning	9-11	Incomplete or incorrect; critical errors or missing components				
☐ Not Attempted	0-8	Indicates failure to perform the task.				
Evaluated by:		Remarks/Comments				
Name of Course Instructor						